# Verified Inspection Desk

- Arabic / English municipal assistance
- The model proposes. The human decides. The database proves it.
- SDAIA Academy capstone | Track 1 | Synthetic data only

# One workflow, end to end

- Verify the permit holder without revealing whether an unverified permit exists.
- Retrieve inspection policy over MCP; find actual district availability.
- Present an exact booking for human approval; commit only after revalidation.
- Demo fixture: FTR-30311 / health inspection / Al-Rawdah.

# Keep authority outside the model

- CLI or authenticated HTTP client -> bounded agent loop
- Public policy MCP -> retrieved rules, no applicant database tools
- Local permit + slot + booking tools -> one persistent session database
- Human approval -> authorization recheck -> atomic write -> read-back

# A prompt is not a permission system

- Four tools only; unknown capabilities are unavailable.
- Identity, current ownership and agent authorization checked in code.
- Denied approval stays denied for that turn; exact slot rechecked at commit.
- Bounded history/results, step and cost limits; redacted evidence.
- Offline verification: 102 checks; full evidence in results/.

# Change one thing; measure honestly

- Suite: 16 original tasks; two configurations.
- A: generated final reply. B: verified booking/denial renderer. Safety unchanged.
- A: 11/16 completed tasks passed (68.8%); 16/16 runs complete
- B: 13/16 completed tasks passed (81.2%); 16/16 runs complete
- Authentic cache replays: A 3, B 48. Real tools, fresh DBs; not independent sampling.
- Full suite complete; scripted synthetic consent, not human sign-off.

# Failures are part of the deliverable

- A loses booking confirmation fields; B fixes two paired outcomes.
- Health denial task: requested slot unavailable; no denial gate exercised.
- Refund answer invents a refund-request capability; no write occurred.
- Arabic needs-info refusal trips a lexical negation false positive.
- Scores and original graders retained; full analysis in docs/EVALUATION.md.
- Policy/refusal prose remains model-generated; rendering is not better planning.
- No pass^3 or production-readiness claim.

# Break a capability chain, not just a phrase

- Private records + untrusted ticket + outbound channel form the trifecta.
- A false audit override requested full identity fields to an attacker address.
- Sensitive outbox rows: 1 unsafe -> 0 defended
- The control is an isolated unsafe lab; the defended run uses our actual app.
- Only a local synthetic outbox is used. No real email. One attack is not comprehensive assurance.

# Reproduce, inspect, question

- uv run rafeeq.py checks
- uv run rafeeq.py chat --session demo
- uv run rafeeq.py serve --host 127.0.0.1 --port 8793
- README + tasks + A/B CSVs + security evidence + slides.
- Limits: synthetic identity, isolated session databases, free-provider availability, heuristic PII masking.